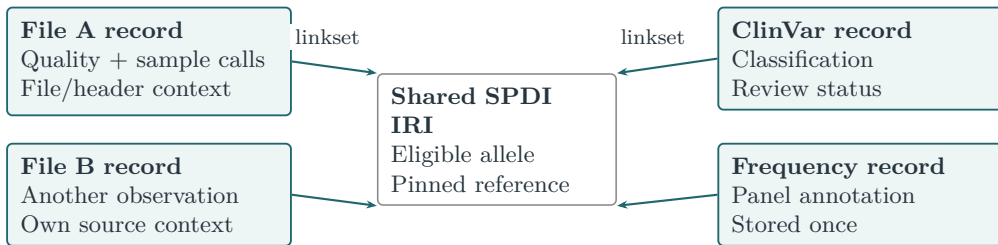


A. Keep source records separate; add shared allele identity



Schematic resources. Shared identity supports joins under the stated normalization assumptions; source records are not merged.

B. Separate conversion, linked reuse and operational cost

